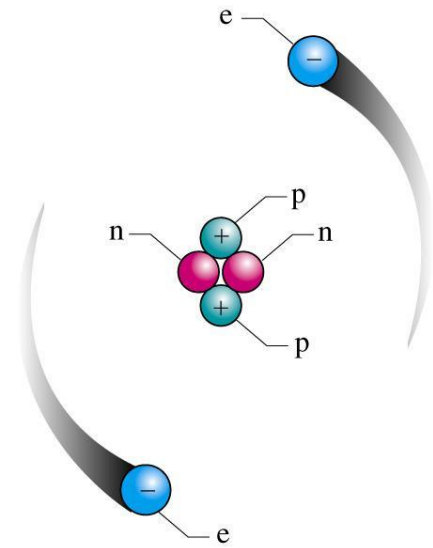


Chemistry

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Matter, Atoms, Molecules and Ions

Chemistry: Chapter 1 and 2

Contents

- Matter, Substance, Mixture, Element & Compound
- Constituents of atom
- Atomic number
- Mass number
- Isotopes
- Molecules and ions
- Mole and Avogadro constant
- Orbital
- Introduction to periodic table

What is Chemistry?

Chemistry is the study of **matter** and the changes it undergoes.

Matter

- Matter is anything that occupies space and has mass.
- Matter includes things we can see and touch (such as water, trees, and earth).
- Matter also includes things we can not see and touch (such as air).

Substance and Mixtures

- A substance is a form of matter that has a fixed (constant) composition and distinct properties.
- Examples: water, sugar, salt, gold and oxygen.
- Substances differ from one another in composition and can be identified by their smell, taste, appearance and other properties.

Mixtures

1 A mixture is a combination of two or more substances in which the substances retain their distinct identities.

2 Example: air, soft drinks, milk, and cement.

3 Mixtures do not have constant composition.

Mixtures

Mixtures are two types:

Homogeneous and heterogeneous

- Homogenous mixture:
- Homogenous mixture is a mixture in which the composition of the mixture is the same throughout.
- When sugar dissolves in water we get a homogeneous mixture.

Mixtures

Heterogeneous mixture:

- Heterogeneous mixture is a mixture where composition is not uniform.
- If sand is mixed with iron filings, the sand grains and the iron filings remain separate. This a heterogeneous mixture.

Elements and Compounds

Substances can be either elements or compounds.

□ Element:

- An element is a substance that cannot be separated into simpler substances by chemical means.
- Chemists use symbols of one or two letters to represent the elements (Example: Al, Fe, Co).

Element

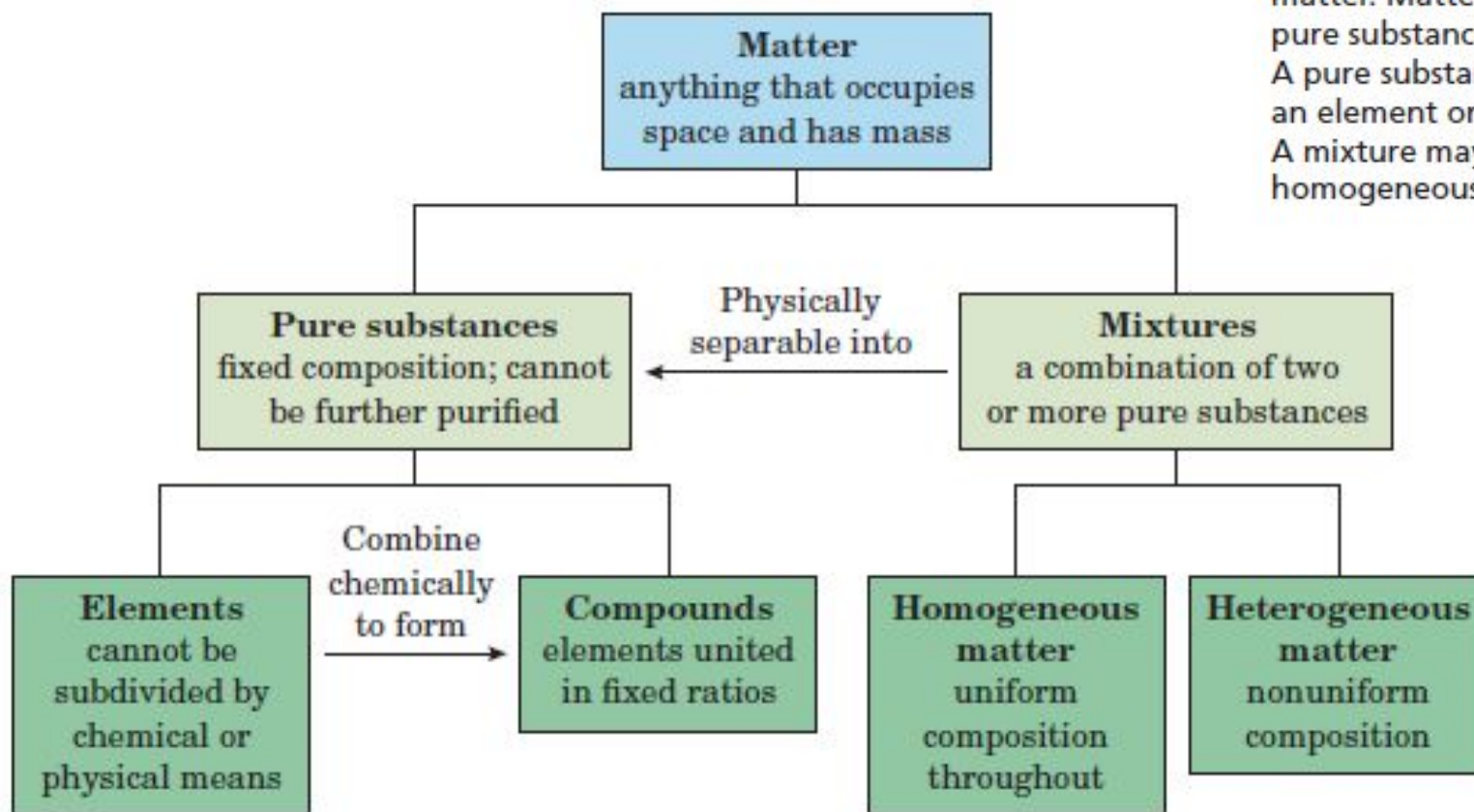
- An element is a substance (for example, carbon, hydrogen, and iron) that consists of identical atoms.
- At this time, 116 elements are known.
- Of these, 88 occur in nature; chemists and physicists have made the others in the laboratory.

Compounds

- Compound:
- A compound is a substance composed of atoms of two or more elements chemically united in fixed proportions.

Matter

FIGURE 2.1 Classification of matter. Matter is divided into pure substances and mixtures. A pure substance may be either an element or a compound. A mixture may be either homogeneous or heterogeneous.



Atom



Human body is made of some 50 trillion to 100 trillion cells

Living things consist of atoms of different **elements**

An element is one type of atom

Elements and their functions in human body

Table 2A Elements and Their Functions in the Human Body

Element	Function	Element	Function
The Big Four		The Trace Elements	
Carbon (C)	The subject of Chapters 10–19 (organic chemistry) and 20–31 (biochemistry)	Chromium (Cr)	Increases effectiveness of insulin
Hydrogen (H)		Cobalt (Co)	Part of vitamin B ₁₂
Nitrogen (N)		Copper (Cu)	Strengthens bones; assists in enzyme activity
Oxygen (O)		Fluorine (F)	Reduces the incidence of dental cavities
The Next Seven		Iodine (I)	An essential part of thyroid hormones
Calcium (Ca)	Strengthens bones and teeth; aids blood clotting	Iron (Fe)	An essential part of some proteins, such as hemoglobin, myoglobin, cytochromes, and FeS proteins
Chlorine (Cl)	Necessary for normal growth and development	Manganese (Mn)	Present in bone-forming enzymes; aids in fat and carbohydrate metabolism
Magnesium (Mg)	Helps nerve and muscle action; present in bones	Molybdenum (Mo)	Helps regulate electrical balance in body fluids
Phosphorus (P)	Present as phosphates in bone, in nucleic acids (DNA and RNA), and involved in energy storage and transfer	Zinc (Zn)	Necessary for the action of certain enzymes
Potassium (K)	Helps regulate electrical balance in body fluids; essential for nerve conduction		
Sulfur (S)	An essential component of proteins		
Sodium (Na)	Helps regulate electrical balance in body fluids		

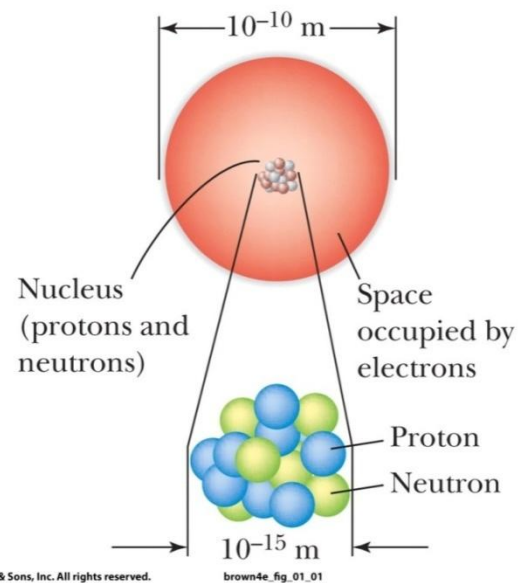
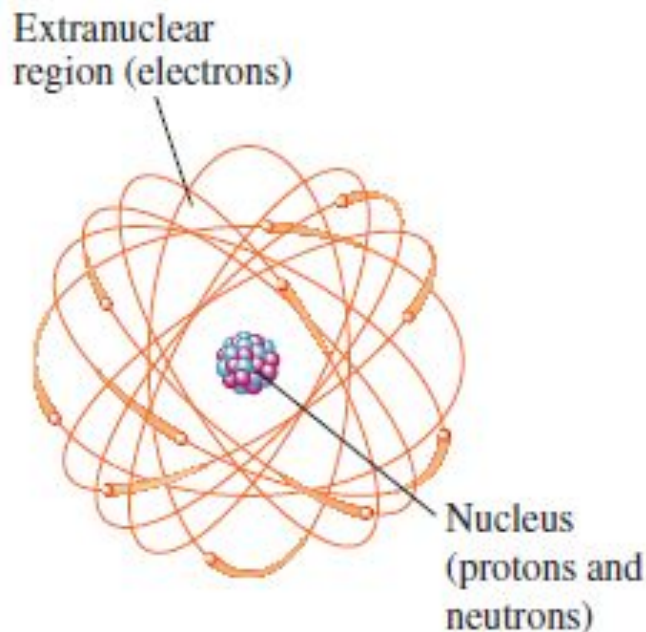
Atom

- An element is one type of atom
- An atom is one of the basic units of matter.
- Everything around us is made up of atoms.
- An atom is a million times smaller than the thickest human hair.

Atom

- ❑ An atom consists of a very dense central nucleus made up of protons and neutrons, plus electrons that move about the nucleus at a relatively large distance from it.
- ❑ Protons are positively charged, neutrons have no charge, and electrons are negatively charged.

Atom



- Diameter of a nucleus is about 10^{-12} cm
- Diameter of an atom: 10^{-8} cm
- Electrons circulate around the nucleus at a distance of approximately 10^{-10} m .

Atom

1. Atoms are made up of *subatomic particles* called electrons, protons, and neutrons.
2. The protons and neutrons form a tiny, dense, central body called the nucleus of the atom.
3. The electrons are distributed in space around the nucleus.

Neutrons are uncharged subatomic particles with nearly the same mass as protons. They are located in the nuclei of atoms.

Comparison of Subatomic Particles

Table 2.1 Properties and Location within Atoms of Protons, Neutrons, and Electrons

Subatomic Particle	Charge	Mass (g)	Mass (amu)	Mass (amu); Rounded to One Significant Figure	Location in an Atom
Proton	+1	1.6726×10^{-24}	1.0073	1	In the nucleus
Electron	-1	9.1094×10^{-28}	5.4858×10^{-4}	0.0005	Outside the nucleus
Neutron	0	1.6749×10^{-24}	1.0087	1	In the nucleus

The mass of a proton is 1.6726×10^{-24} g, but this number is so small that it is more convenient to use another unit, called the **atomic mass unit (amu)**.

$$1 \text{ amu} = 1.6605 \times 10^{-24} \text{ g}$$

Problem

Q1. Mass of a carbon atom is 1.99×10^{-26} kg. How many carbon atoms are present in a sample of carbon of mass 10 g?

Solve

Q1. Mass of a carbon atom is 1.99×10^{-26} kg. How many carbon atoms are present in a sample of carbon of mass 10 g?

Solve:

$$\begin{aligned}\text{Number of carbon atoms} &= \frac{\text{Mass of sample}}{\text{Mass of one atom}} \\ &= \frac{1.00 \times 10^{-2} \text{ kg}}{1.99 \times 10^{-26} \text{ kg}} \\ &= 5.03 \times 10^{23}\end{aligned}$$

Scale of Atoms

Useful units:

★ $1 \text{ amu (atomic mass unit)} = 1.66054 \times 10^{-24} \text{ kg}$

★ $1 \text{ pm (picometer)} = 1 \times 10^{-12} \text{ m}$

★ $1 \text{ Å (Angstrom)} = 1 \times 10^{-10} \text{ m} = 100 \text{ pm} = 1 \times 10^{-8} \text{ cm}$

$$1 \text{ amu} = 1.6605 \times 10^{-24} \text{ g}$$

Atomic number and mass number

Atomic number (Z):

Number of protons in the nucleus of each atom of an element is called the atomic number (Z).

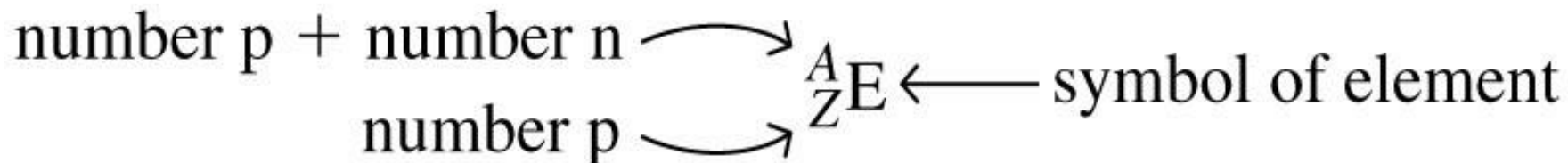
Atomic number (Z) = No. of protons in atom's nucleus
= No. of electrons around atom's nucleus

Mass number (A):

Mass number (A) is the total number of neutrons and protons present in the nucleus of an atom of an element.

Atomic number and mass number

□ To represent a particular atom we use the symbolism:



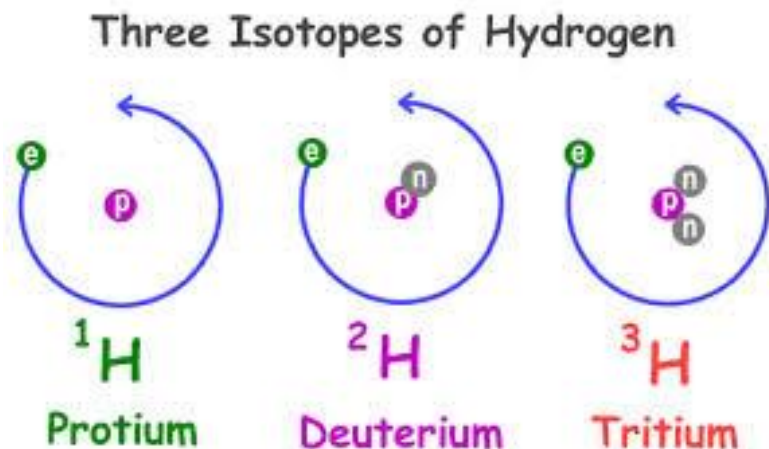
A = mass number Z = atomic number

Mass number (A) = the number of protons + neutrons
in the nucleus of an atom

Isotope

Atoms with identical atomic numbers but different mass numbers are called isotopes.

Hydrogen has three isotopes.



(Non radioactive) (Non radioactive) (Radioactive)

